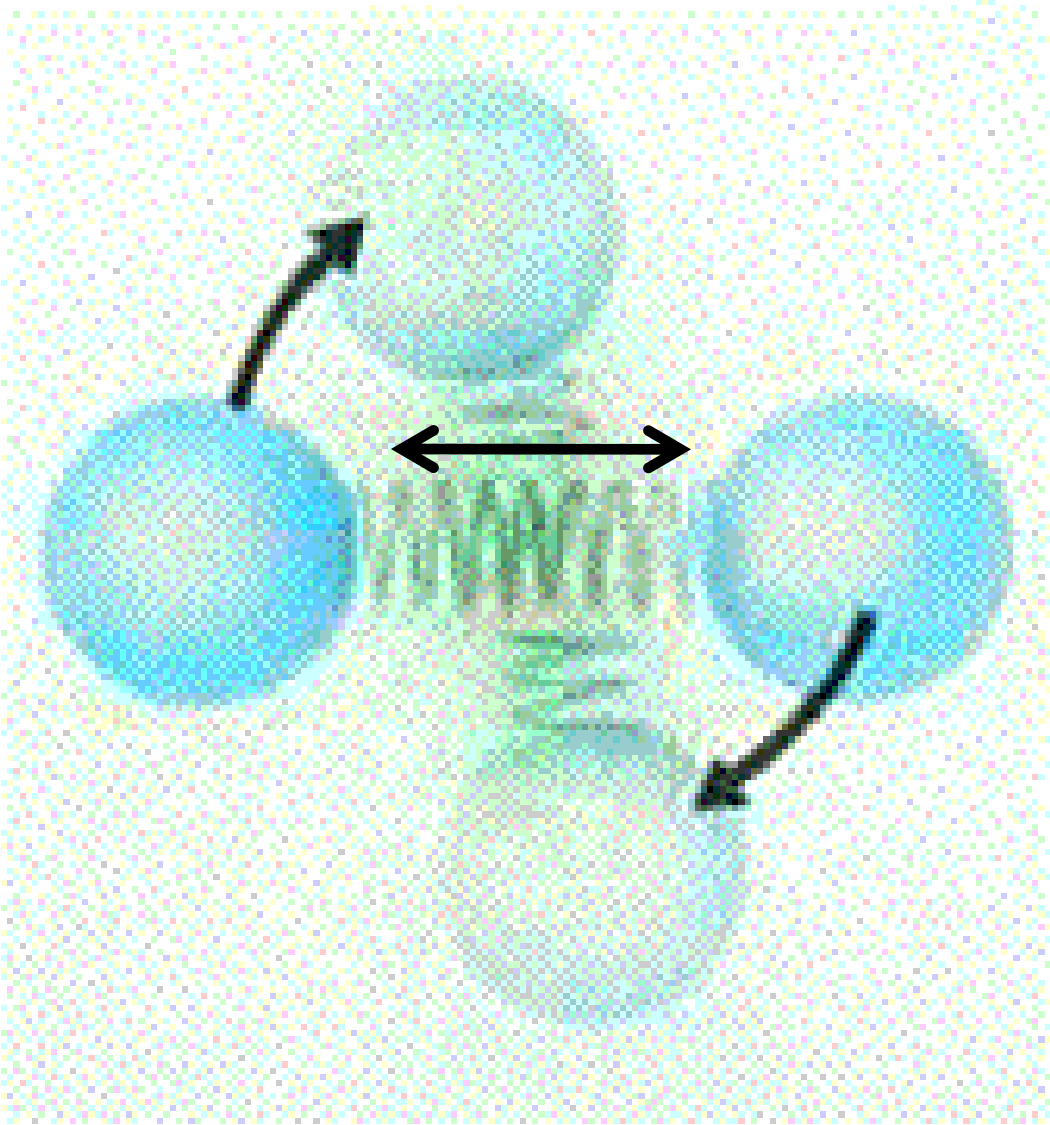
The Diatomic Vibrating Rotator



Approximation:

A diatomic molecule can execute rotations and vibrations quite independently (<u>considering the respective energy gaps</u>).

Born-Oppenheimer Approximation.

The combined rotational-vibrational energy is simply the sum of the separate energies.

$E_{total} = E_{rot.} + E_{vib.}$ (joules)

$\varepsilon_{total} = \varepsilon_{rot.} + \varepsilon_{vib.}$ (cm^{-1})

$$\varepsilon_{total} = \varepsilon_{rot.} + \varepsilon_{vib.} \qquad (cm^{-1})$$

Hence,

$$\varepsilon_{total} = \varepsilon_{J,\upsilon} = \varepsilon_J + \varepsilon_\upsilon = BJ(J+1) - DJ^2(J+1)^2 + HJ^3(J+1)^3 + \ldots..$$

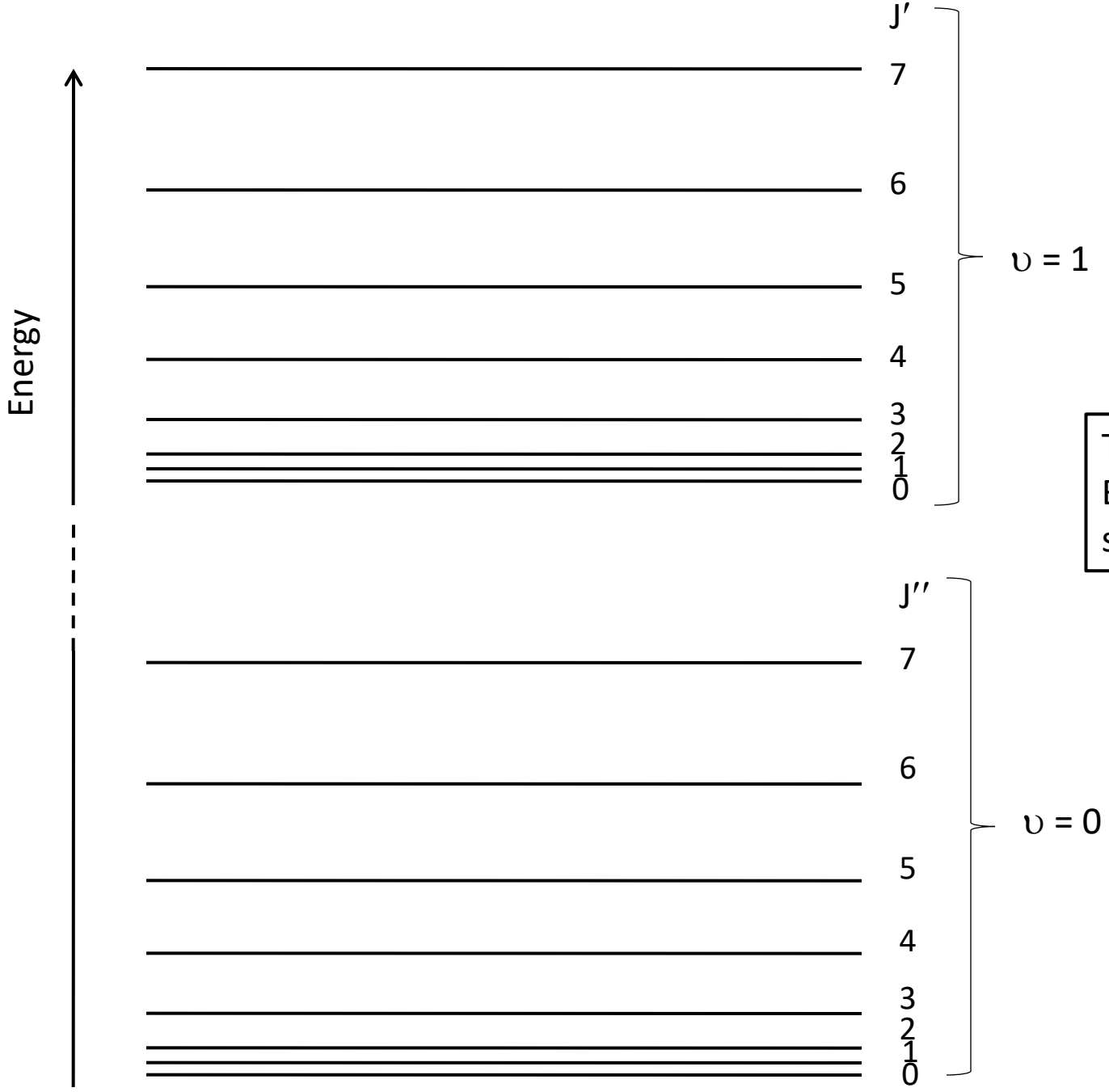
$$+\left(\upsilon + \frac{1}{2}\right)\overline{\omega}_e - x_e\left(\upsilon + \frac{1}{2}\right)^2 \overline{\omega}_e \ \ \mathrm{cm}^{-1}$$

Ignoring the small centrifugal distortion constants D,H, etc.

$$\varepsilon_{J,\upsilon} = BJ(J+1) + \left(\upsilon + \frac{1}{2}\right)\overline{\omega}_e - x_e\left(\upsilon + \frac{1}{2}\right)^2 \overline{\omega}_e \ \ \mathrm{cm}^{-1}$$

??? The molecule is rigid yet vibrating!!! **(not logical)**

D has a very minor effect on the spectrum.



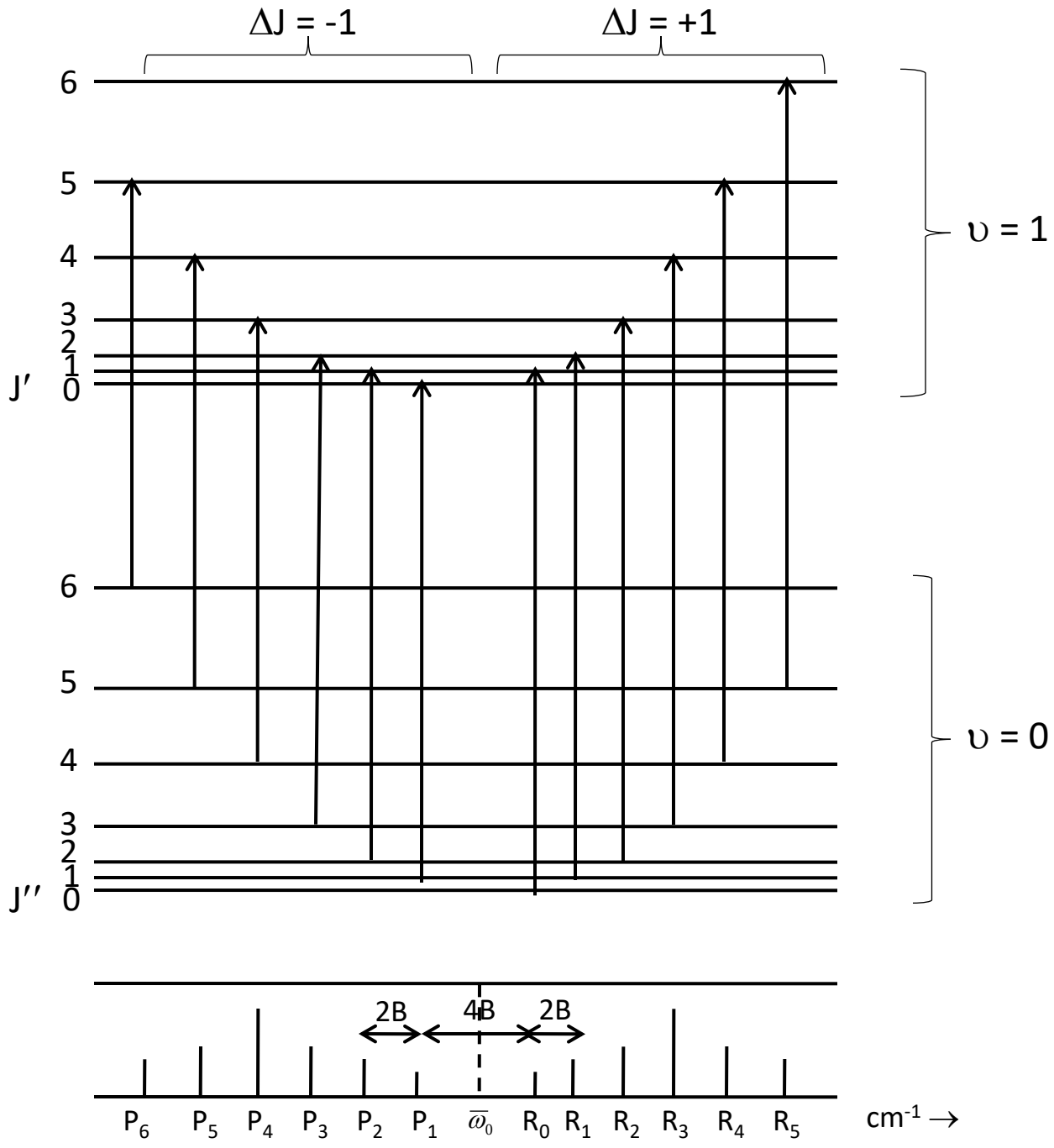
The rotational constant B is taken to be the same for all J and v.

Selection Rules:

$\Delta\upsilon = \pm 1, \pm 2$, **etc.** $\Delta J = \pm 1$

A vibrational change must be accompanied with a rotational change.

$\Delta\upsilon = 0$ will mean a purely rotational transition.
$\Delta J = 0$ may not be feasible for a diatomic rotor unless there is a very special case.



Filled to varying degrees in molecular population (Boltzmann distribution).

Considering the $\upsilon = 0 \rightarrow \upsilon = 1$ transition,

$$\Delta\varepsilon_{J,\upsilon} = \varepsilon_{J',\upsilon=1} - \varepsilon_{J'',\upsilon=0}$$

$$= BJ'(J'+1) + 1\frac{1}{2}\overline{\omega}_e - 2\frac{1}{4}x_e\overline{\omega}_e - \{BJ''(J''+1) + \frac{1}{2}\overline{\omega}_e - \frac{1}{4}x_e\overline{\omega}_e\}$$

$$= \overline{\omega}_0 + B(J' - J'')(J' + J'' + 1) \ \text{cm}^{-1}$$

where, $\overline{\omega}_0 = \overline{\omega}_e(1 - 2x_e)$

According to Born-Oppenheimer approximation

<u>B is identical in the upper and lower states</u> $\rightarrow$ Rotation is unaffected by molecular vibration.

So,

$\Delta J = +1 \Rightarrow J'-J'' = +1$, hence,

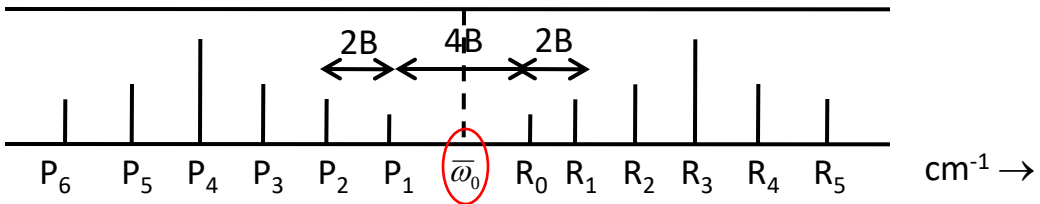
$$\Delta\varepsilon_{J,\upsilon} = \overline{\omega}_0 + 2B(J''+1)\ \text{cm}^{-1} \qquad J'' = 0,1,2,\ldots$$

Similarly,

$\Delta J = -1 \Rightarrow J'-J'' = -1$, hence,

$$\Delta\varepsilon_{J,\upsilon} = \overline{\omega}_0 - 2B(J'+1)\ \text{cm}^{-1} \qquad J' = 0,1,2,\ldots$$

$\overline{\omega}_0$ is the "band origin" or "band centre".



In general,

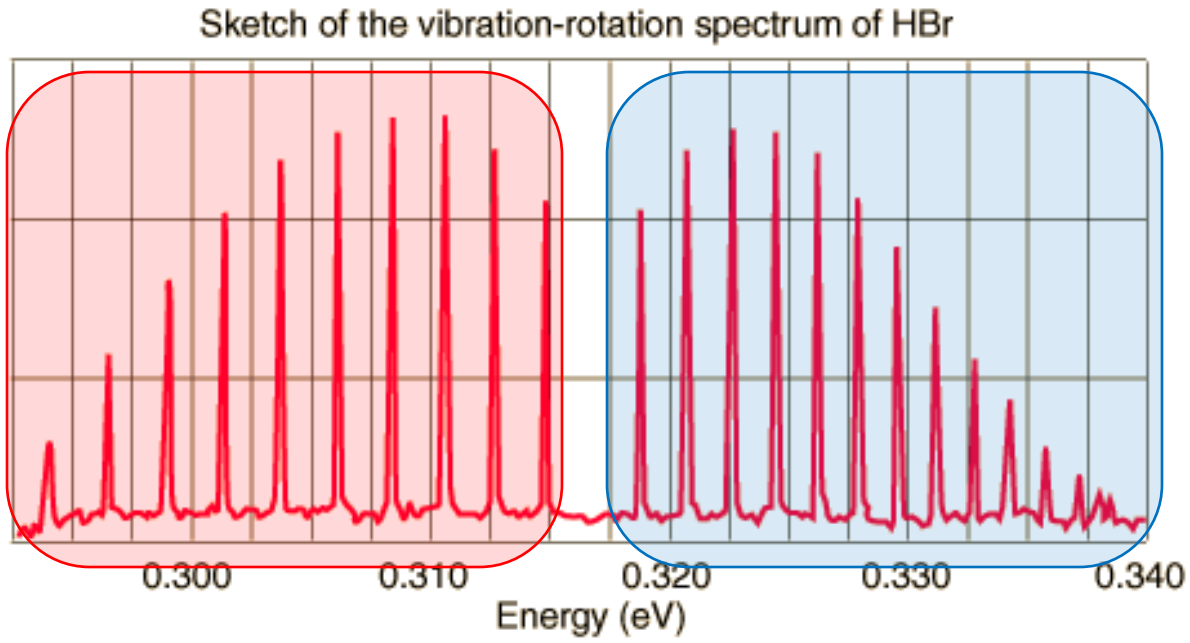
$$\Delta\varepsilon_{J,\upsilon} = \overline{\nu}_{spect.} = \overline{\omega}_0 + 2Bm \ \ \mathrm{cm}^{-1} \qquad\qquad m = \pm 1, \pm 2, \ldots$$

m cannot be 0 $\rightarrow$ then J′ or J′′ will be -1.

This equation represents combined vibration-rotation spectrum.

The spectrum will consist of equally spaced lines (spacing = 2B) on both sides of the band origin.

The line at the band origin will not appear since $m \neq 0$.

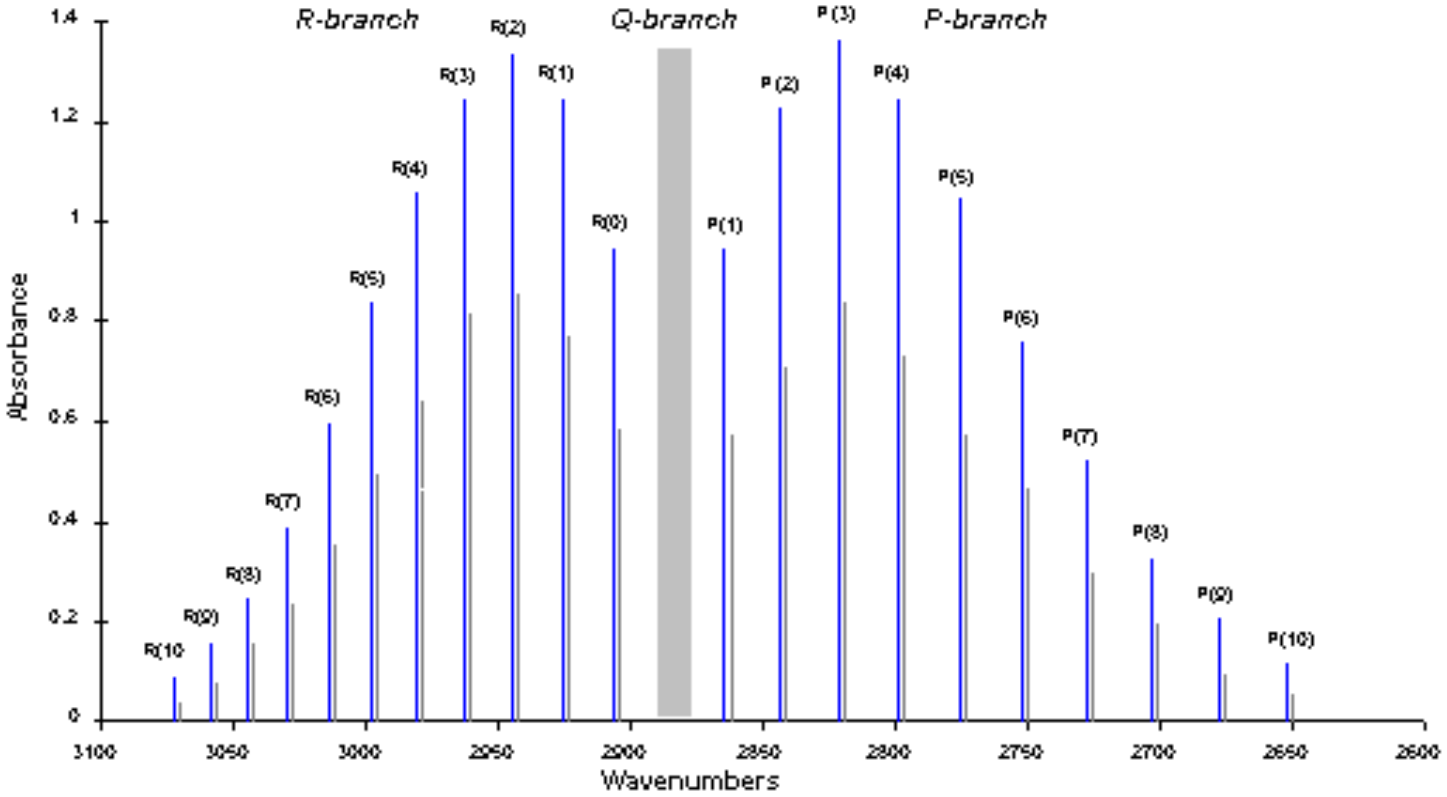


Lines to the low frequency side of the band origin corresponding to negative *m* ($\Delta J = -1$) $\rightarrow$ ***P branch***.

Lines to the high frequency side of the band origin corresponding to positive *m* ($\Delta J = +1$) $\rightarrow$ ***R branch***.

Lines from $\Delta J =$	-2	-1	0	+1	+2	
Are called	***O***	***P***	***Q***	***R***	***S***	branch

Isotope Effect



The effect of the reduced mass on $\nu_{harmonic}$ (disregarding anharmonicity)

$$\frac{\nu^*_{harmonic}}{\nu_{harmonic}} = \sqrt{\frac{\mu}{\mu^*}}$$

For the effect of different isotopes on rotation

$$\frac{B_e^*}{B_e} = \frac{\mu}{\mu^*}$$

How does rotation influence the spectra of polyatomic molecules?

Selection Rule: $\Delta\upsilon = \pm1, \pm2, \pm3, \ldots.$ $\Delta J = \pm1$ $\Delta J \neq 0$

In order to be IR active, there must be a dipole change during the vibration:

either along the major axis of rotational symmetry or parallel (||) to it

<u>or</u> at right angles to the line (perpendicular, ⊥).

The selection rules for the rotational transitions of complex molecules depend on the <u>type of vibration</u>, || or ⊥, which the molecule is undergoing.

The selection rules also depend on the shape of the molecule.

Linear molecules

Parallel Vibrations

Selection Rules:

$\Delta J = \pm 1$, $\Delta \upsilon = \pm 1$ (for simple harmonic oscillators)
$\Delta J = \pm 1$, $\Delta \upsilon = \pm 1, \pm 2, \pm 3, \ldots$ (for anharmonic oscillators)

The spectra will contain the P and R branches and no line occurring at the band centre.

For moderately big linear molecules like HCN or CO_2

the moment of inertia may be considerably larger,
the B value considerably smaller,
the P or R line spacing may be less.

For still larger linear molecules

the B value will be so small that the P or R branches may not be clearly differentiated.

Linear molecules

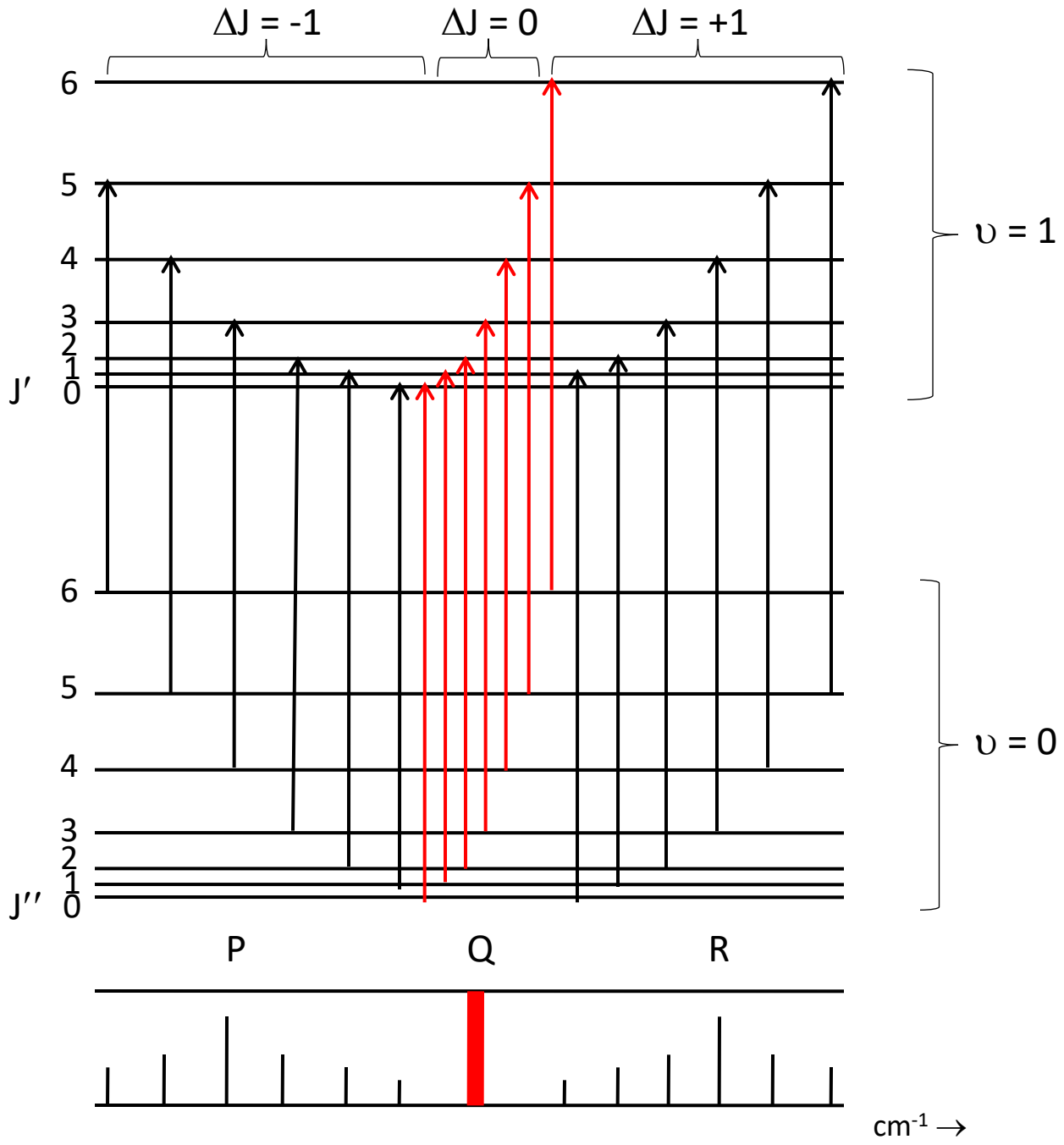
Perpendicular Vibrations

Selection Rules:

$\Delta J = 0, \pm 1, \Delta \upsilon = \pm 1$ (for simple harmonic oscillators)

$\Downarrow$

A vibrational change can take place with no simultaneous rotational transition.


ΔJ = -1
ΔJ = 0
ΔJ = +1
6
5
4
3
2
1
J′ 0
υ = 1
6
5
4
3
2
1
J′′ 0
υ = 0
P
Q
R
cm-1 →

For Q branch

$$\Delta\varepsilon = \varepsilon_{J,\upsilon=1} - \varepsilon_{J,\upsilon=0}$$

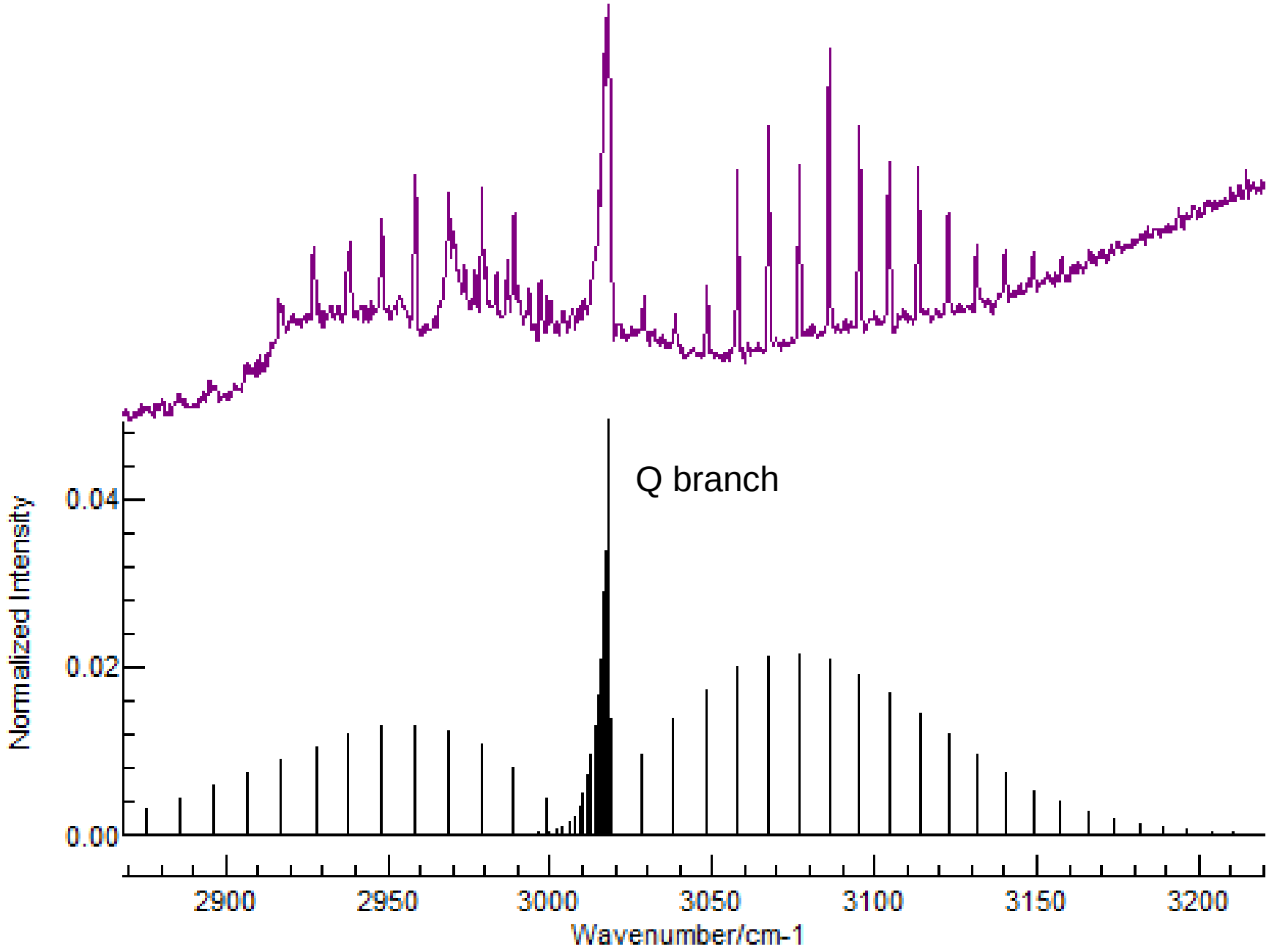
$$= 1\frac{1}{2}\overline{\omega}_e - 2\frac{1}{4}x_e\overline{\omega}_e + BJ(J+1) - \left\{\frac{1}{2}\overline{\omega}_e - \frac{1}{4}x_e\overline{\omega}_e + BJ(J+1)\right\}$$

$$= \overline{\omega}_e(1-2x_e) = \overline{\omega}_0 \text{ cm}^{-1} \qquad \text{for all J}$$

The Q branch contains lines superimposed upon each other at the band centre.

The resultant line is very intense.

The Q branch whether present or not corresponds to a vibrational transition ONLY. All the other bands are due to vibrational and rotational changes occurring at the same time.


Q branch
Normalized Intensity
0.04
0.02
0.00
2900
2950
3000
3050
3100
3150
3200
Wavenumber/cm-1